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RESEARCH PAPER

TMPO-AS1-hsa-let-7b-5p-EZH2-RNA network predicts poor survival in basal-like breast cancer patients

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ABSTRACT

Background: The study explores the role of non-coding ribonucleic acids (ncRNAs) and higher enhancer of zeste homolog 2 (EZH2) gene in breast cancer progression, examining how microRNA (miRNA) and long noncoding RNA (lncRNA) control EZH2, potentially influencing oncogene growth and treatment failure.

Materials and methods: The databases used in the study included Cyclebase 3.0 and CellTracer to determine EZH2's role in cell cycle, Oncomine, OncoMX and The University of Alabama At Birmingham Cancer Data Analysis Portal (UALCAN) for Pan-cancer analysis, The Cancer Genomic Atlas Portal (TCGA Portal), Gene Expression Profiling Interactive Analysis (GEPIA2), OncoDB, CR2Cancer, Encyclopaedia of RNA Interactomes (ENCORI), and The Cancer Genome Altas Analyzer (TCGAnalyzeR v1.0) for differential expression analysis, CR2Cancer, OncoDB, MethMarkerDB, and Wanderer databases for epigenetic alteration analysis, Kaplan-Meier Plotter for survival analysis, Breast Cancer Gene Expression Miner (bc-GenExMiner v5.0) for hormone receptor analysis, Tumor-Immune System Interaction Database (TISIDB), Cancer Single Cell State Atlas (CancerSEA), TNMplot, DriverDBv4, and ENCORI for biological processes, cell cycle checkpoints and metastasis analysis, Enrichr, Tumor Immune Estimation Resource (TIMER 2.0), Gepia2 for transcription factor analysis, miRNet, Transcriptome Alterations in Cancer Omnibus (TACCO), and CancerMIRome for miRNA analysis, Enrichr, UALCAN, and ENCORI for lncRNA analysis.

Results: The EZH2 gene is overexpressed in breast cancer (BRCA) tumors, metastatic tissues, and circulating tumor cells, potentially leading to cancer progression. Patients with high EZH2 expression have shorter overall survival (OS), distant metastasis-free survival (DMFS), and relapse-free survival (RFS) compared to those with low expression. Estrogen receptot (ER)-negative BRCA tumors and PR-negative tumors have high EZH2 gene and eukaryotic transcription factor (E2F2) levels. The EZH2/E2F2 axis may assist ER/PR-negative BRCA by sponging homosapiens microRNA family (hsa-let-7b-5p) through lncRNA-thymopoietin antisense transcript 1 (TMPO-AS1). Overexpression of the EZH2 protein is associated with BRCA metastasis.

Conclusion: EZH2 overexpression in basal-like BRCA is mediated by a competing endogenous RNA (ceRNA) network and regulating their expression levels may facilitate better survival outcomes.

Keywords: breast cancer; ceRNA network; EZH2; Kaplan-Meier plotter; prognosis

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Introduction

Breast cancer is a global disease with the highest incidence rate among all cancers. In 2020, it had an estimated 2.3 million new cases, representing 11.7% of all new cancers [1]. Prognostic and predictive factors for future recurrence or death from breast cancer include patient age, comorbidity,

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tumor size, tumor grade, number of involved axillary lymph nodes and, possibly, a biomarker status [2, 3]. Algorithms have been published to estimate recurrence rates, and a validated computer-based model (Adjuvant! Online for breast cancer) is available to estimate 10-year disease-free survival [4]. Breast cancer detection and intervention in the early stages are key to improving prognoses and reducing mortality rates. Researchers have used various diagnostic approaches, including mammography, magnetic resonance imaging, ultrasound, biopsies, serum screening for micro ribonucleic acid (miRNAs), blood-based proteomics, biomarker analysis, and biosensor technologies [5]. Comprehensive treatment primarily involves surgery combined with chemotherapy, endocrine therapy, radiation therapy, and targeted therapies [6, 7]. New gene-expression profiling studies have found four different types of breast cancer: the basal-like type, the human epidermal growth factor receptor 2 (HER2)-negative type, and the two luminal estrogen receptor (ER)-positive types, which have been named luminal A and luminal B [8]. These newly defined molecular subgroups have distinct clinical outcomes. Gene-expression profiling technology aims to provide better predictions of clinical outcomes than traditional clinical and pathological parameters [9].

Studies have shown that enhancer of zeste homolog 2 (EZH2) is frequently overexpressed in breast cancer cells, silencing tumor suppressor genes and promoting tumor cell growth and poor survival [10]. Targeting a competing endogenous miRNA–long noncoding RNA (lncRNA)–EZH2-RNA network could be an effective strategy for treating breast cancer. High levels of EZH2 expression are associated with a poorer prognosis for several types of cancer, including breast cancer [11], prostate cancer [12], and bladder cancer [13]. In breast cancer, patients with high EZH2 expression have a higher risk of recurrence and shorter overall survival. Clinical trials are developing and testing new drugs targeting EZH2, providing a fresh approach to cancer treatment [14–16]. This study of the miRNA-lncRNA-EZH2-RNA network and its function in breast cancer will lead to the development of novel medicines that will improve the prognosis for individuals suffering from this dreadful illness. This article focuses on the role of hypomethylation in cancer development, particularly human breast cancer. It investigates how miRNA, lncRNA, and gene overexpression of EZH2 impact oncogene development and resistance to treatment.

Materials and methods

Expression analysis of EZH2

For this study, we used Cyclebase 3.0 [17], CellTracer [18], Oncomine [19], OncoMX [20] and The University of Alabama At Birmingham Cancer Data Analysis Portal (UALCAN) [21] to assess EZH2 expression levels in the cell cycle and in tumor tissue *vs.* normal tissues across pan-cancer. The study examined how EZH2 mRNA expression differed in breast cancer (BRCA) *vs.* normal samples. It was achieved by utilizing databases such as UALCAN, The Cancer Genomic Atlas Portal (TCGA Portal) [22], Gene Expression Profiling Interactive Analysis (GEPIA2) [23], OncoDB [24], CR2Cancer [25], Encyclopaedia of RNA Interactomes (ENCORI) [26], and The Cancer Genome Altas Analyzer (TCGAnalyzer v1.0) [27]. We analyzed the methylation status of EZH2 in BRCA using the UALCAN, CR2Cancer, OncoDB, MethMarkerDB [28], and TCGA Wanderer [29] databases. The study used Breast Cancer Gene-Expression Miner v5.0 [30] to look at the connection between EZH2 expression levels and the patients' clinical and pathological features, such as their estrogen, progesterone receptor status, and UALCAN and Tumor-Immune System Interaction Database (TISIDB) [31] databases for BRCA subtypes and expression of EZH2 based on patient's age analysis. Using Cancer Single Cell State Atlas (CancerSEA) [32], we investigated the association of EZH2 with BRCA's biological process. ENCORI database was used to determine the correlation between EZH2 and Cyclin-CDKs checkpoints. The metastatic analysis was carried out using TNMplot [33] and DriverDBv4 [34].

The survival analysis of EZH2 and its transcription factors

The study used the BRCA datasets and the Kaplan-Meier Plotter [35] to look at relapse-free survival (RFS), overall survival (OS), distant metastasis-free survival (DMFS), and post progression survival (PPS). It did this by dividing patients by median and looking at the results. The screening conditions were as follows: "Cancer: BRCA"; "Gene symbol: EZH2"; Affy id: 203358_s_at. Using the Enrichr [36] database, we examined transcriptional factors involved in tissue-specific gene expression regulation. The study also analyzed the correlation between the eukaryotic transcription factor (E2F) family and the EZH2 gene using the ENCORI, Tumor Immune Estimation Resource (TIMER) [37], OncoDB, and GEPIA2 databases.

An analysis of non-coding-related regulatory networks

The study analyzed the miRNA network associated with EZH2 using the miRNet [38] database. We verified the correlation between EZH2 and miRNA using the ENCORI, Transcriptome Alterations in Cancer Omnibus (TACCO) [39], and CancerMIRNome [40] databases. We used the CancerMIRNome, Kaplan-Meier (KM) plotter, and ENCORI to figure out what the EZH2-associated miRNA meant for prognosis. The CancerMIRNome and UALCAN were used to look at differences in miRNA expression based on normal *vs.* tumor expression. We used the Enrichr and UALCAN databases to identify the lncRNA associated with EZH2. Using the ENCORI, GEPIA2, OncoDB, and miRNet databases, we confirmed the co-relationship between EZH2 and specific lncRNA. We also determined individual lncRNA expression using UALCAN and OncoDB.

A molecular mechanism analysis of EZH2

The study validated the relationship between a gene and ESR1, PGR, using the ENCORI, OncoDB, and Breast Cancer Gene Expression Miner (bc-GenExMiner v5.0) databases. It also identified a correlation between the competing endogenous RNA (ceRNA) network and ESR1 and PGR, including E2F2, homosapiens microRNA family (hsa-let-7b-5p), and thymopoietin antisense transcript 1 (TMPO-AS1).

Statistical analysis

The study analyzed EZH2 gene expression differences between tumor and normal tissues, using log-rank tests to compare survival, performance heterogeneity, and gene enrichment, with statistical significance at $p < 0.05$. We analyzed the validity of the data using statistical methods based on online databases.

Results

EZH2 expression levels in pan-cancer and breast cancer

To find out what role EZH2 plays in the cell cycle and cancer progression, we first used Cyclebase 3.0 to examine EZH2 expression in the phases of cell cycle. As shown in Figure 1A, EZH2's peak time is in the S phase. Next, we used the CellTracer database, which provided extensive insights into the impact of various relationships on cell development pathways and served as a foundation for identifying biomarkers and investigating novel treatments within the tumor environment. We found that EZH2 is significantly associated with quiescence, invasion, and hypoxia, which are some of the crucial hallmarks of cancer, but its expression is not involved in apoptosis, which helps in breast cancer progression, as shown in Figure 1B. The TISIDB database was also used to identify the role of EZH2 in various biological processes, molecular functions, cellular components, KEGG, and reactomes, as listed in Supplementary File — Tab. S1. Using the Oncomine and OncoMX databases, we were able to show that all TCGA tumors had the heightened levels of EZH2. The levels were higher in tumor tissue than in normal tissue (Fig. 1C and Supplementary File — Tab. S2). In addition, we validated the pan-cancer view using the UALCAN database, and we found the same results as mentioned above, as shown in Figure 1D. The above-mentioned results strongly indicate the oncogenic properties of EZH2 and hence its role in tumor progression. Next, statistical analysis revealed a high expression of EZH2 in breast cancer compared to the healthy group. This difference was statistically significant ($p < 0.05$), as shown in Figures 2A–F. The databases used were UALCAN, TCGA Portal, GEPIA2, oncoDB, CR2Cancer, and ENCORI. Transcriptomic analysis was also carried out in breast invasive carcinoma cells using TCGAnalyzeR v1.0, as shown in Figure 2G, and the data showed upregulation in the EZH2 expression level in BRCA with a log-fold change of 2.08. We linked the amount of methylation in the promoter region to tumor formation. As a result, we investigated EZH2 promoter methylation in pan-can using the CR2Cancer database (Supplementary File — Tab. S3) and found a strong association of EZH2 methylation with breast cancer tissues as compared to other TCGA

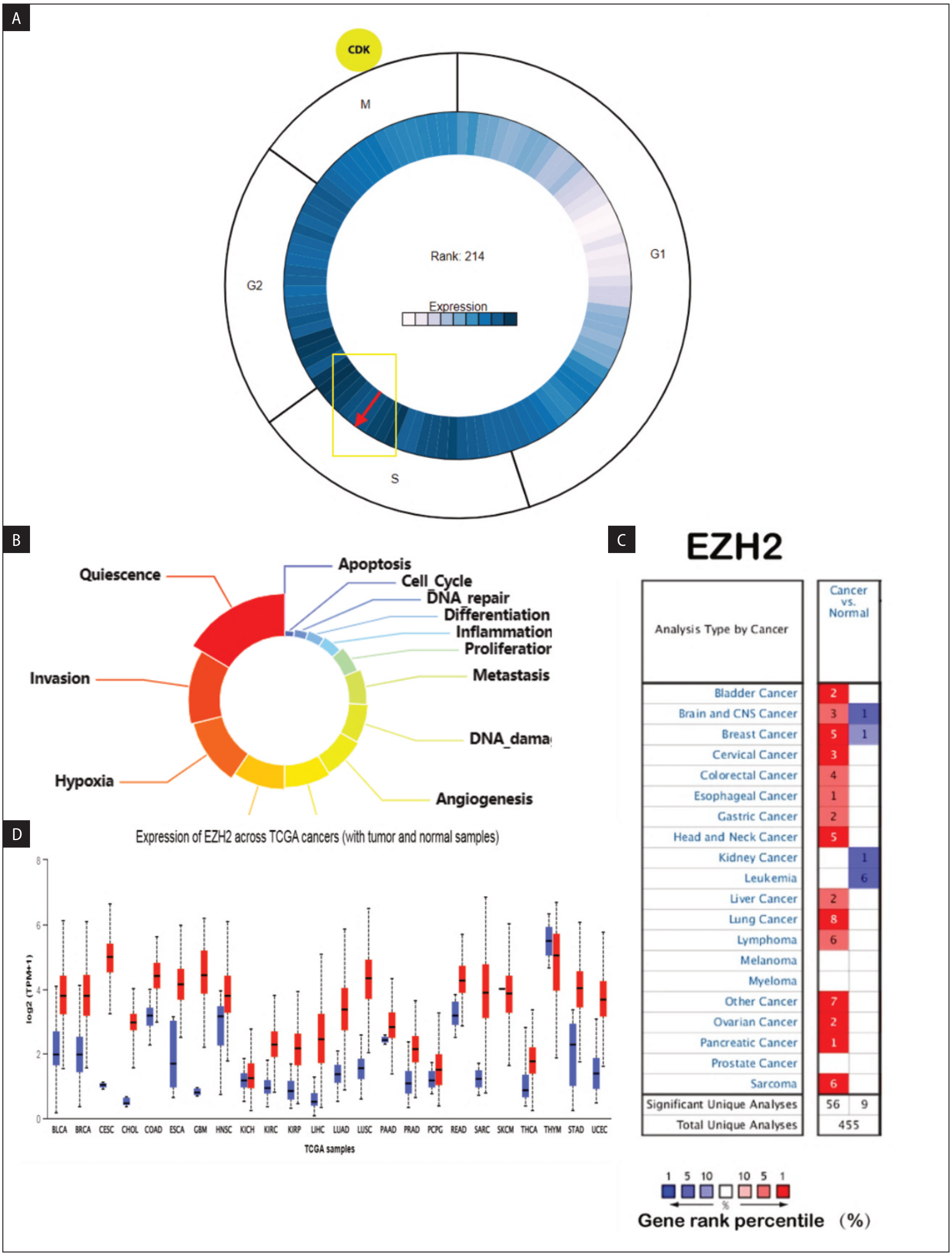

Figure 1. Expression level of EZH2 by using: **A.** Cyclebase 3.0 shows the participation of EZH2 in S-phase in cell cycle; **B.** Involvement of EZH2 in various cell development pathways using Cell Tracer; **C.** Expression of EZH2 in various cancer using Oncomine; **D.** Expression profile of enhancer of zeste homolog 2 (EZH2) was determined by the University of Alabama At Birmingham Cancer Data Analysis Portal (UALCAN) database for tumor versus normal samples; Red bars = Tumors, blue bars corresponding normal tissue

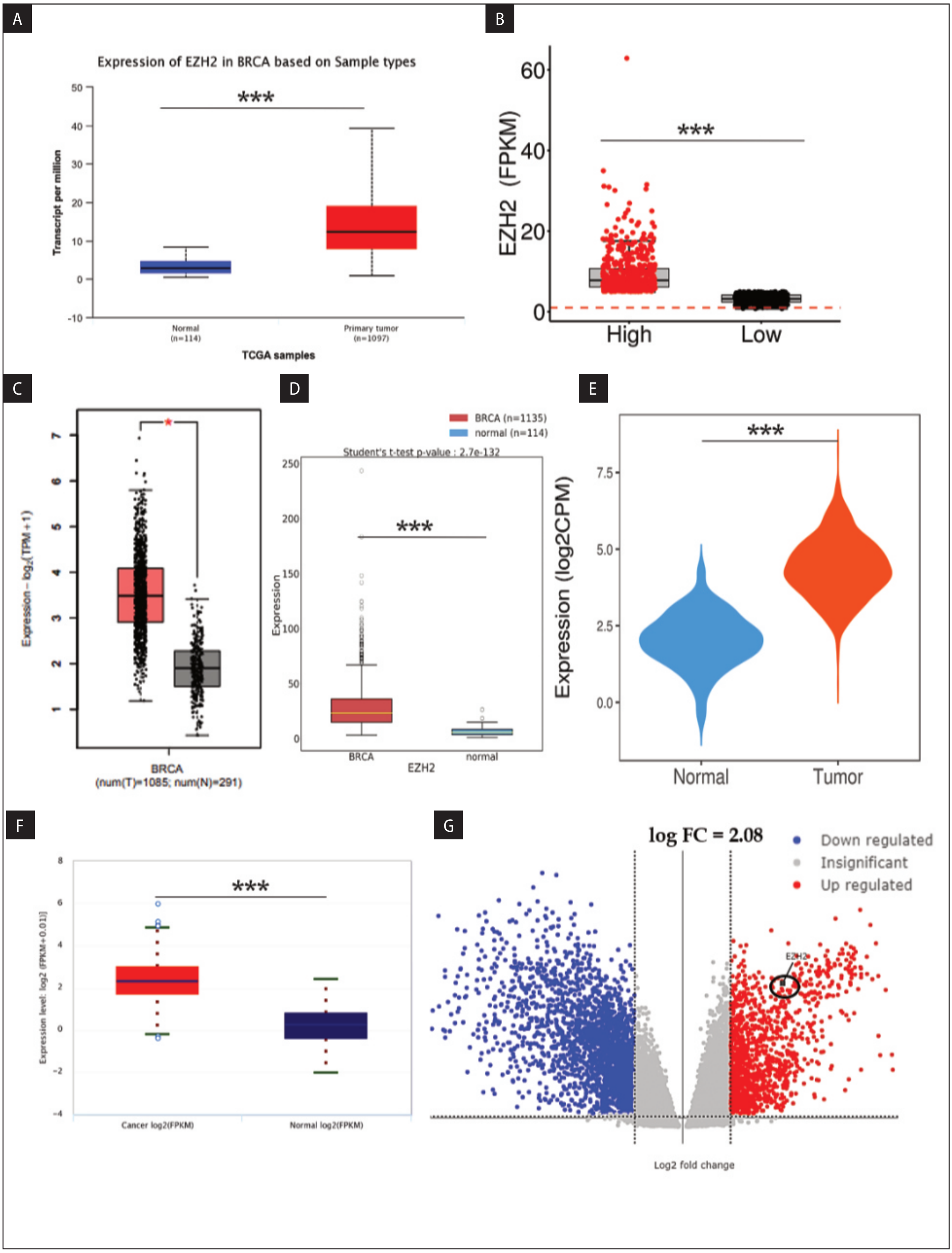

Figure 2. Expression of EZH2 in breast cancer (**A–F**) mRNA expression was analyzed in normal breast tissue and primary tumors from the publicly available. **A.** UALCAN (Normal n = 114, Tumor n = 1097); **B.** TCGA Portal; **C.** GEPIA 2 (Normal n = 291, Tumor n = 1085); **D.** OncoDB; **E.** CR2cancer; **F.** ENCORI; **G.** Transcriptome analysis

cancers, (R = –0.277, p = 1.27e–16). The negative correlation value indicated towards the hypomethylation status of EZH2. Further, the databases used to analyze the methylation profile were UALCAN, CR2Cancer, OncoDB, and MethMarkerDB, respectively. Figure 3A–E reveals that the EZH2 promoter methylation level in breast cancer was much lower than in normal tissues, indicating hypomethylation. The findings indicate a connection between the degree of methylation in the promoter

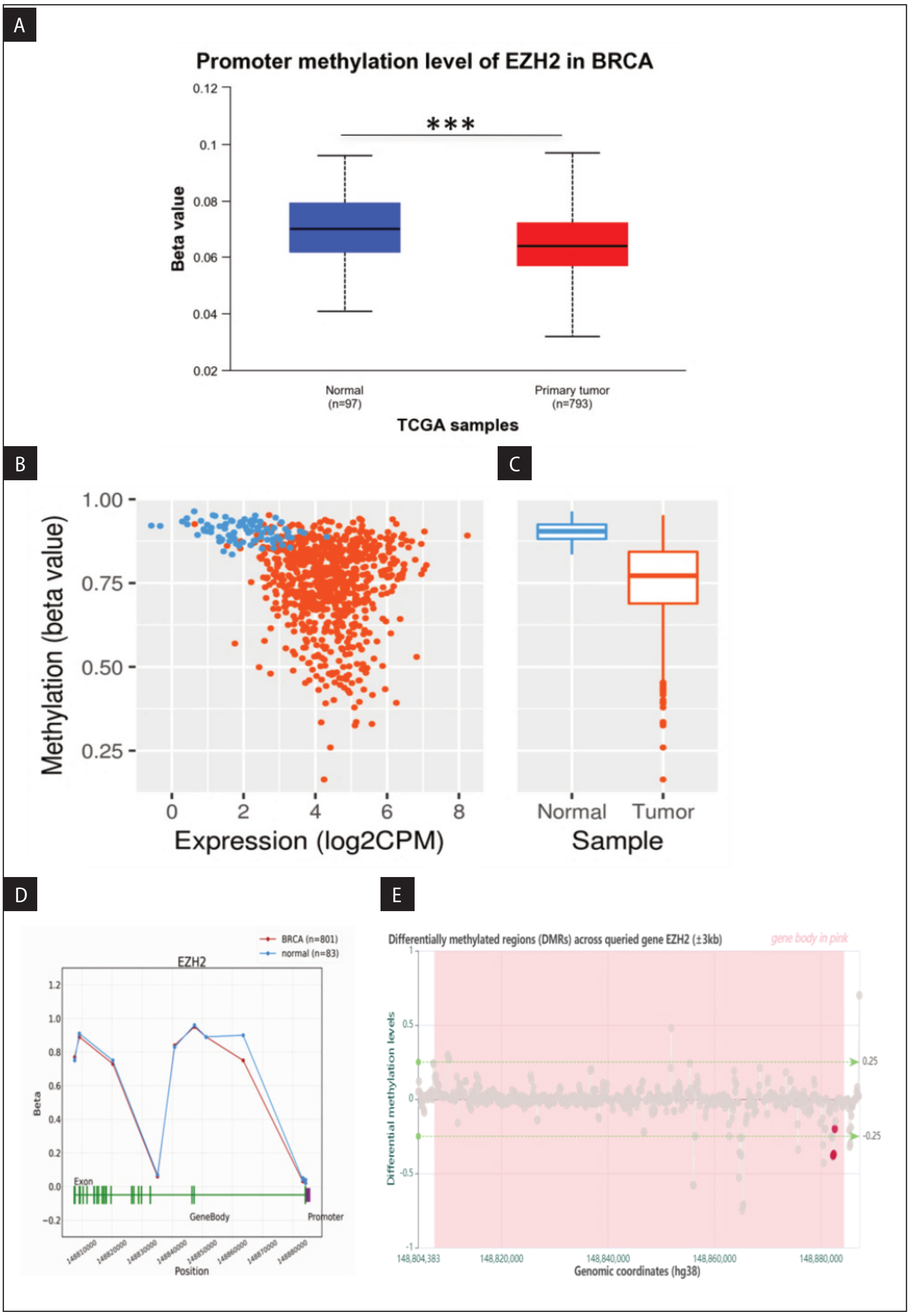

Figure 3. Enhancer of zeste homolog 2 (EZH2) involvement in the methylation using: **A.** University of Alabama At Birmingham Cancer Data Analysis Portal (UALCAN) (Normal n = 114, Tumor n = 1097); **B.–C.** CR2cancer (Normal *vs.* Tumor); **D.** OncoDB; **E.** MethMarkerDB

region and EZH2 expression, which, in turn, influences the survival status of breast cancer patients. Also, Wanderer, an interactive tool to explore deoxyribonucleic acid (DNA) methylation and gene expression data in human cancer, was used to compare the methylation status of EZH2 in a (n = 30) normal sample *vs.* (n = 30) BRCA tumor, and the results showed hypomethylation in breast invasive carcinoma patients, as shown in Supplementary File — Fig. 1A–C.

EZH2 expression level survival analysis in breast cancer

The KM Plotter online database provided the dataset from which we analyzed the EZH2 gene in 4,929 patients. In the 120-month study, higher levels of EZH2 transcripts were linked to worsening RFS, OS, DMFS, and PPS for all breast cancer patients. These were RFS [hazard ratio (HR) = 1.5, confidence interval (CI) = 136–167, p = 6.2e–15], OS (HR = 1.31, CI = 1.08–158, p = 0.0063), DMFS (HR = 1.45, CI = 1.24–1.07, p = 3.9e-6), and PPS (HR = 1.47, CI = 1.16–186, p = 0.0012), as shown in Figure 4A–D. Also, as listed in Supplementary File — Tab. S4, the difference between the high expression cohort and low expression cohort suggests that targeting and inhibiting the levels of EZH2 in BRCA patients can significantly result in good prognosis improving the survival outcomes. By using gold standards like marker of proliferation Ki-67 (MKI67), ESR1, and HER2 biomarkers, we looked more closely at the EZH2 expression multivariate analysis and found a strong link between it and these markers, as shown in Figure 4E (HR = 1.5, CI = 1.36–1.67, P = 6.2e–15). We concluded from this overall analysis that there is a significant correlation between EZH2 gene expression and the poor prognosis of breast cancer.

EZH2 expression with ER and PR receptors and breast cancer subclass

We also aimed to confirm the association between EZH2 expression and Estrogen-Progesterone receptors using Breast Cancer Gene-Expression Miner v5.0. Our findings revealed an overexpression level of ER and PR negative hormone receptors, as depicted in Figure 5A–C. This suggests a strong correlation between EZH2 expression and ER/PR negative breast cancer. We were very curious about EZH2's role in the breast cancer subclass following the validation gene expression and correlation study findings. Using three distinct databases, namely bc-GenExMiner v5.0, UALCAN, and TISIDB, our analysis revealed a significant enhancement of EZH2 in triple negative breast cancer (TNBC) and basal subclasses, as illustrated in Figure 5D–G. Also, by using the bc-GenExMiner v5.0 TCGA RNA seq dataset, we identified the expression level of EZH2 in healthy, tumor-adjacent, non-basal-like and non-TNBC, and basal-like and TNBC, and, interestingly, the results showed the highest expression of EZH2 in Basal-like and TNBC as compared to others, suggesting an important role of EZH2 in aggressive tumors as shown in Figure 5H. After this study, we used the ENCORI database to look at the EZH2 gene in relation to the ESR1 and PGR genes. To our surprise, we discovered that the EZH2 gene had a strong negative correlation with the ESR1 (R = –0.425, p = 1.01e–49) and PGR (R = –0.399, p = 1.59e–43) genes. Figure S2A–B (Supplementary File) demonstrates a close link between EZH2 overexpression and aggressive types of breast cancer. Further, we analyzed EZH2's expression based on patient's age group and the results revealed that overexpression of EZH2 is more aggressive at a younger age as compared to the older patients as shown in Figure 5I.

EZH2 metastasis and biological processes involvements

We also analyzed EZH2 in relation to various cellular biology processes and found a significant positive correlation between EZH2 and the cell cycle (0.56), DNA repair (0.49), DNA damage (0.43), and cell proliferation (0.36), with a p-value of less than 0.05, as illustrated in Figure 6A–E. As established earlier, EZH2 plays an important role in cancer progression, which aligns with our analysis of cell cycle checkpoints as listed in Supplementary File — Tab. S5. EZH2 showed a strong positive correlation with Cyclin E1/CDK2 (R = 0.67 and 0.57, respectively) and Cyclin B1/CDK1 (R = 0.67 and 0.69, respectively), key regulators of the S and M phases of the cell cycle. Whereas, the weaker correlation between G0/G1 and G2 phases checkpoints reflect a more specialized function of EZH2 in cell division and tumor progression. After that,

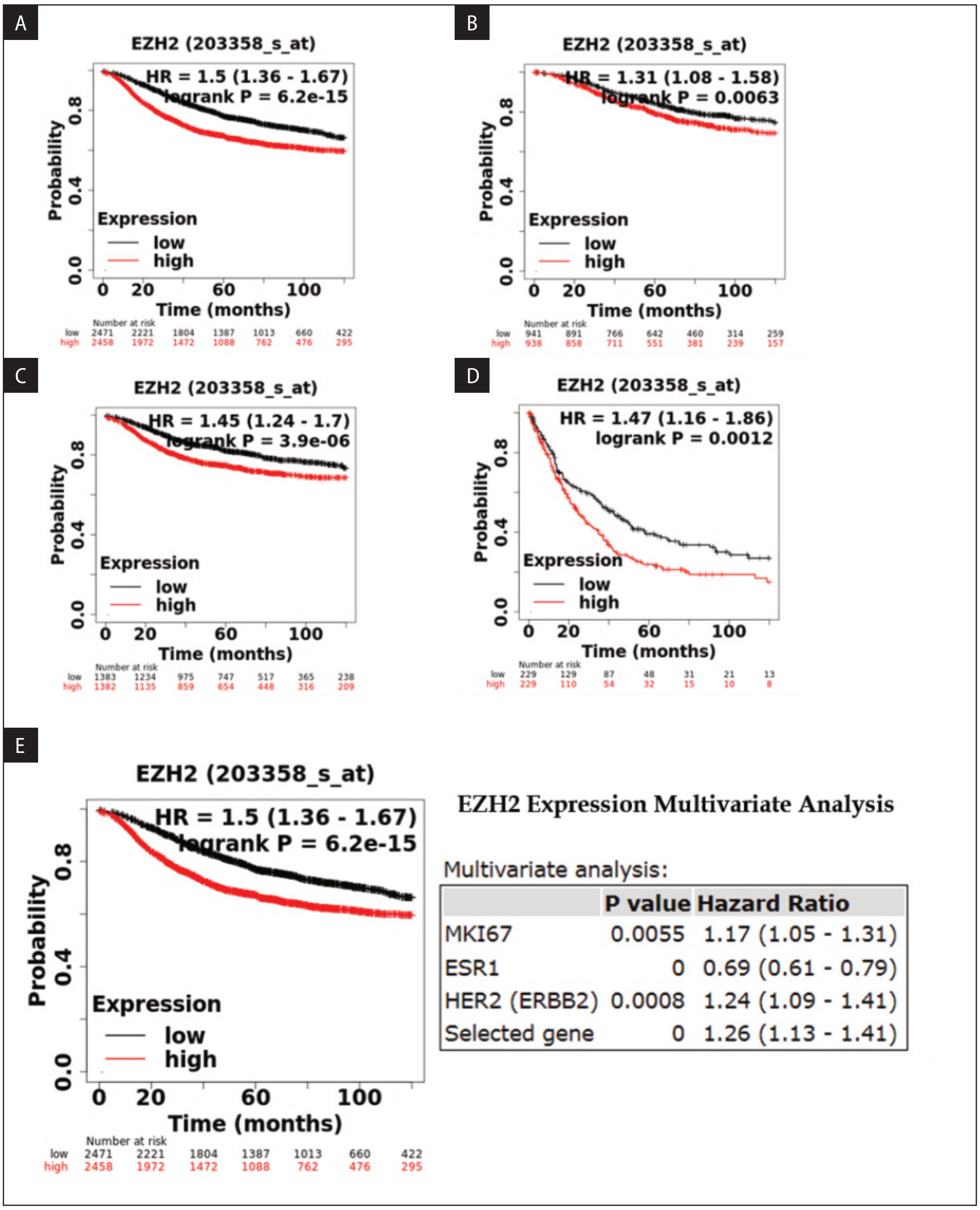

Figure 4 A.–E. Enhancer of zeste homolog 2 (EZH2) survival analysis. Kaplan-Meier survival curves were plotted for: **A.** Relapse-free survival (RFS); **B.** Overall survival (OS); **C.** Distant metastasis-free survival (DMFS). **D.** Post progression survival (PPS); **(E)** Multivariate analysis

we did a metastasis analysis with the TNMplot database to examine the connections between the EZH2 genes, and we found significant metastasis with gene chip (p = 1.58 e–28, as shown in Figure 6F) and with RNA-Seq (p = 5.01 e–53, as shown in Figure 6G), and the data was later corroborated with DriverDBv4 databases, revealing a considerably increased level of EZH2 in the metastatic phase, as seen in Figure 6H.

EZH2 transcriptional regulation in BRCA

Cancer cells highly depend on the constitutive expression of transcriptional factors (TFs) to support their growth and survival. Increasing knowledge of the TFs' mechanisms of action and regulatory networks has led to a better understanding of their roles in cancer and other diseases. Given the critical role TFs play in cancer, researchers have made significant efforts to develop drugs that target

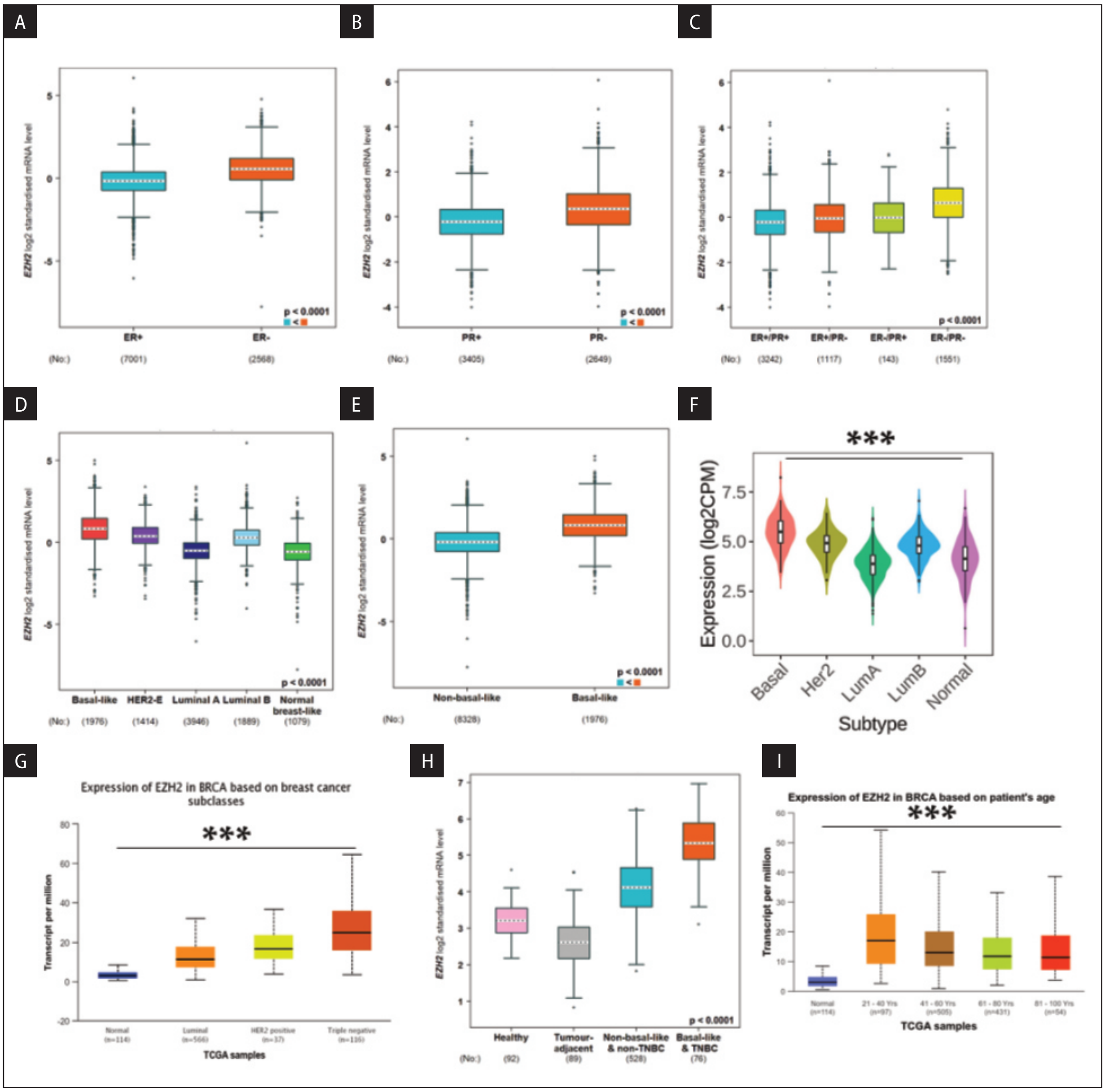

Figure 5. **A.–E.** mRNA expression of ESPL 1 in breast cancer aggressiveness and subtypes by using bc-GenExMiner v5.0; **F.** Enhancer of zeste homolog 2 (EZH2) gene expression with the breast cancer subtypes using Tumor-Immune System Interaction Database (TISIDB); **G.** University of Alabama At Birmingham Cancer Data Analysis Portal (UALCAN); **H.** By using bc-GenExMiner v5.0 database (Healthy *vs*. Tumor adjacent *vs*. BRCA subtypes); **I.** Expression analysis based on patient's age group using UALCAN

TFs. Several drugs targeting TFs in cancer are currently in different phases of clinical trials. We used the Enrichr database (Supplementary File — Tab. S6) to identify transcriptional factors associated with EZH2, as well as the ENCORI database to identify correlations. We then chose eight transcriptional factors from the E2F family, including E2F1, E2F2, E2F3, E2F4, E2F5, E2F6, E2F7, and E2F8, as shown in Figure 7 A–D and Supplementary File — Fig. S2C–F, and found that out of those eight, only four transcriptional factors (E2F1, E2F2, E2F7, and E2F8) have the highest correlation value with EZH2, which states that the activity or expression level of the transcription factor is closely related to the activity or expression level of the EZH2 gene. To validate it further, we used three different databases, such as TIMER, OncoDB, and the GEPIA database, as shown in Figure 7E–P. After this study,

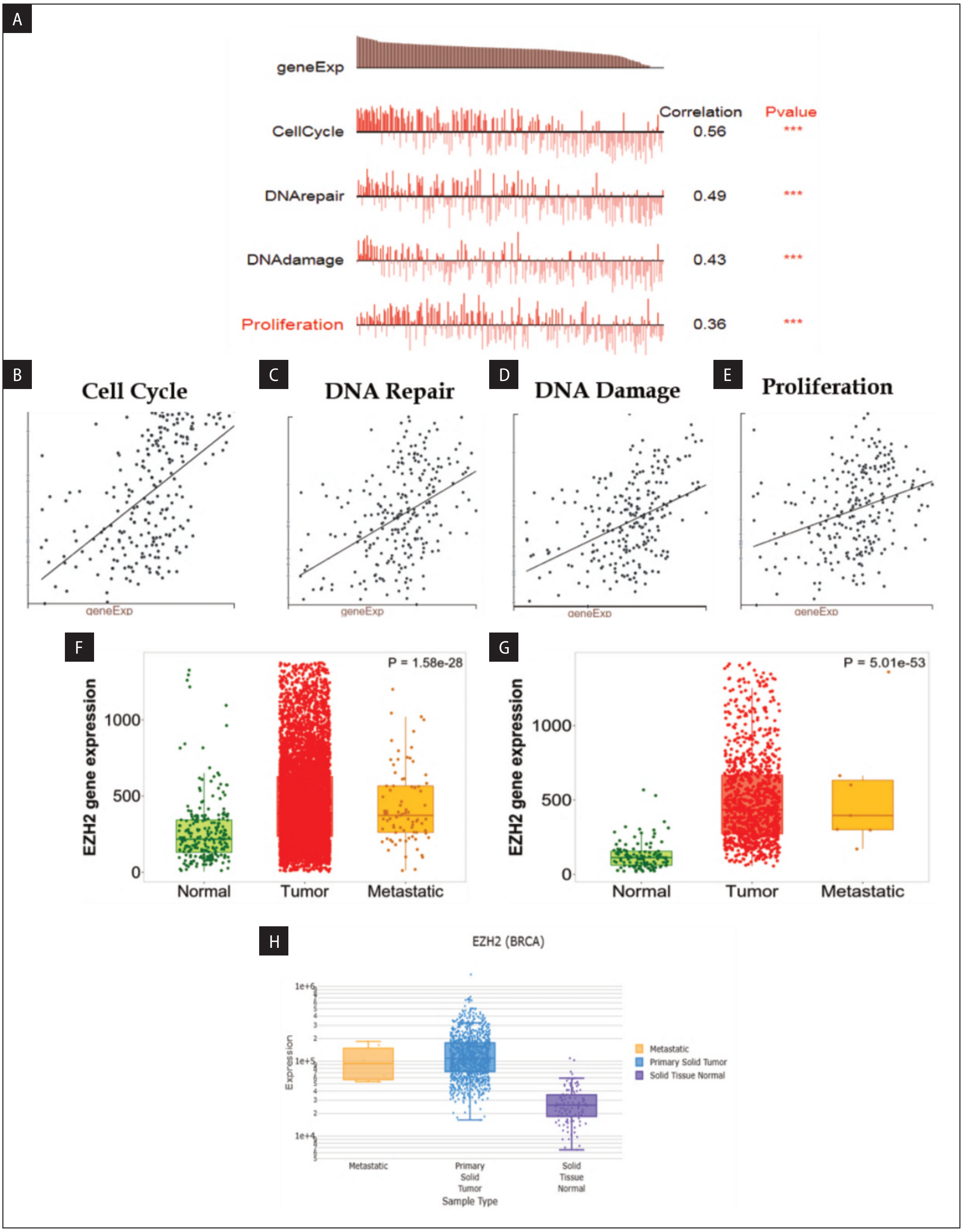

Figure 6. Enhancer of zeste homolog 2 (EZH2) expression in tumors from breast cancer patients with biological process and metastasis. **A.–E.** EZH2 expression in biological process using CancerSEA; **F.–H.** Metastasis in EZH2 expression normal, tumor, and metastasis in breast cancer patients using the Gene Chip and RNA — Chip using Tumor Node Metastasis Plot (TNM plot) and DriverDB database

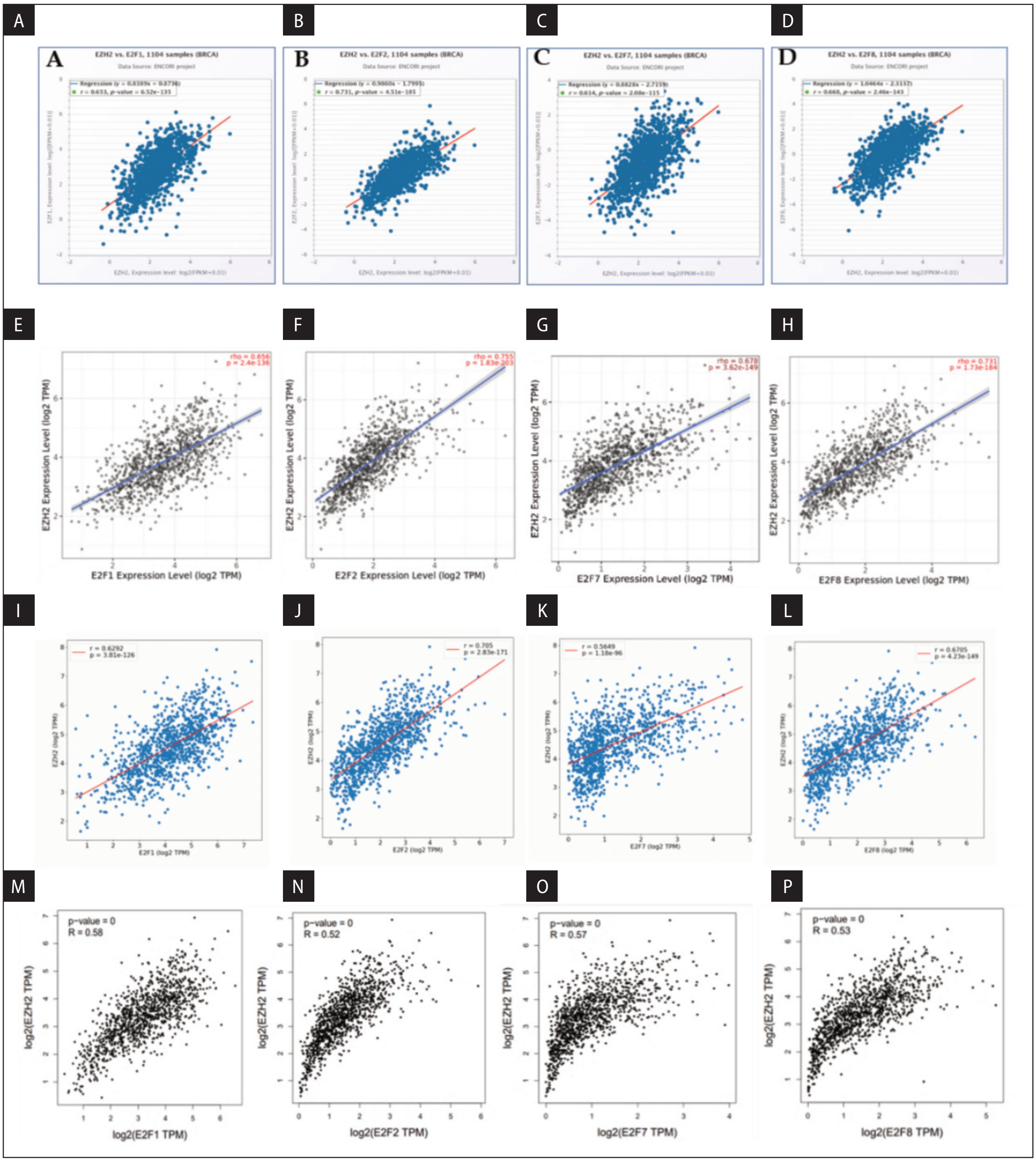

Figure 7. A.–D. Correlation between the enhancer of zeste homolog 2 (EZH2) and eukaryotic transcription factor (E2Fs) using Encyclopaedia of RNA Interactomes (ENCORI) database; **E.–H.** EZH2 and E2Fs family expression using Tumor Immune Estimation Resource (TIMER) database; **I.–L.** OcoBD database; **M.–P.** GEPIA2.0 database

we used the ENCORI database to look at the E2F1, E2F2, E2F7, and E2F8 genes in relation to the ESR1 and PGR genes. To our surprise, we discovered that the E2F2 gene had the highest negative correlation with the ESR1 and PGR genes (–0.409 and –0.395, respectively). The Supplementary File — Fig. S3A–H demonstrates a close link between E2F1, E2F2, E2F7, and E2F8 overexpression and aggressive types of breast cancer.

EZH2-non-coding RNA regulatory network

The mechanism for BRCA's EZH2 overexpression is unclear. Supplementary File — Fig. S4A shows the top 24 miRNAs associated with EZH2 identified by the miRNet database, a platform for visualizing miRNA networks. Further study using the ENCORI database revealed that only 7 miRNAs (hsa-let-7b-5p, hsa-mir-26a-5p, hsa-mir-101-3p, hsa-mir-214-3p, hsa-mir-217, hsa-mir-1-3p, and hsa-miR-101-3p) out of 24 miRNAs exhibited a negative correlation with EZH2 gene expression, as presented in Table 1. The table also showed that hsa-let-7b-5p has the best negative correlation (R = –0.289) with EZH2 in BRCA patients. To ensure the reliability of our findings, we utilized the TACCO, CancerMIRNome, and ENCORI databases. And we found that EZH2 and hsa-let-7b-5p negatively correlated with each other, as shown in Figure 8A–C. Next, we analyzed the hsa-let-7b-5p expression in BRCA patients compared to normal by using the CancerMIRNome database, and we found that hsa-let-7b-5p was downregulated in BRCA patients as compared with normal, as shown in Figure 8D. This down-expression was further validated by using the UALCAN database, and we found similar results as shown in Figure 8E. Survival analysis of hsa-let-7b-5p in BRCA patients using the CancerMIRNome (HR = 0.58, CI = 0.42–0.80, p = 8.37e–04), KM plotter (HR = 0.68, CI = 0.56-0.83, p = 0.00014), and ENCORI (HR = 0.56, p = 0.00038) databases showed that the downregulation of hsa-let-7b-5p is associated with poor prognosis, as shown in Supplementary File — Fig. S4B–D. Also, the correlation analysis of hsa-let-7b-5p with ESR1 and PGR genes using the ENCORI database (Supplementary File — Fig. S4E–F) showed a significant relationship between hsa-let-7b-5p *vs.* ESR1 (R = 0.267, p = 3.60e–19) and hsa-let-7b-5p *vs.* PGR (R = 0.274, p = 3.64e–20), indicating a positive association with the expression of hsa-let-7b-5p and ESR1/PGR genes. This study also investigates how EZH2, miRNAs, and lncRNAs work together. Using the Enrichr database, we discovered several lncRNAs linked to EZH2. The lncRNAs shown in Supplementary File — Tab. S7 are TMPO-AS1, PRC1-AS1, RRM1-AS1, H2AZ1-DT, LINC01775, SGO1-AS1, DEPDC1-AS1, HMMR-AS1, UBL7-AS1, and APOBEC3B-AS1. We checked how each lncRNA was expressed in the case of BRCA using UALCAN and found that 4 out of 10 lncRNAs were significantly upregulated, as shown in Supplementary File — Tab. S8. Further, we correlated these lncRNAs to EZH2 in the BRCA using ENCORI as shown in Table 2 and found that TMPO-AS1 was strongly positively correlated with EZH2 (R = 0.587, P = 4.16e–103, shown in Figure 8F) as compared to others. Further validation of the data was done using the GEPIA2, and OncoDB databases, Figures 8G–H show that TMPO-AS1 had the strongest positive correlation (R = 0.45, R = 0.5014 respectively) with EZH2. We also found that TMPO-AS1 is significantly overexpressed in BRCA patients compared to normal. We used the UALCAN and OncoDB databases, as shown in Figure 8I–J. Further correlation between E2F2 *vs.* TMPO-AS1 showed a strong positive correlation between both, with an R value of 0.561 and a P value of 1.46e-92. However, on analyzing the correlation between hsa-let-7b-5p and TMPO-AS1, a negative correlation was observed (R = –0.204, p = 1.12e–11). It shows that TMPO-AS1 has a sponge effect on hsa-let-7b-5p

Table 1. Correlation of enhancer of zeste homolog 2 (EZH2) *vs.* micro ribonucleic acid (miRNAs)

S.no	miRNA	Correlation
1	hsa-let-7b-5p	–0.289
2	hsa-let-7e-5p	0.031
3	hsa-mir-26a-5p	–0.091
4	hsa-mir-27a-3p	0.174
5	hsa-mir-101-3p	–0.16
6	hsa-mir-214-3p	–0.072
7	hsa-mir-217	–0.013
8	hsa-mir-200b-3p	0.164
9	hsa-mir-137	0.100
10	hsa-mir-150-5p	0.182
11	hsa-mir-429	0.215
12	hsa-mir-19a-3p	0.461
13	hsa-mir-27b-3p	0.087
14	hsa-mir-29a-3p	0.009
15	hsa-mir-29b-3p	0.055
16	hsa-mir-29c-3p	–0.262
17	hsa-mir-34a-5p	0.042
18	hsa-mir-1-3p	–0.159
19	hsa-mir-155-5p	0.334
20	hsa-mir-16-5p	0.267
21	hsa-mir-7-5p	Positive
22	hsa-miR-101-3p	0.305
23	hsa-mir-30d-5p	0.035
24	hsa-mir-27a-3p	0.174

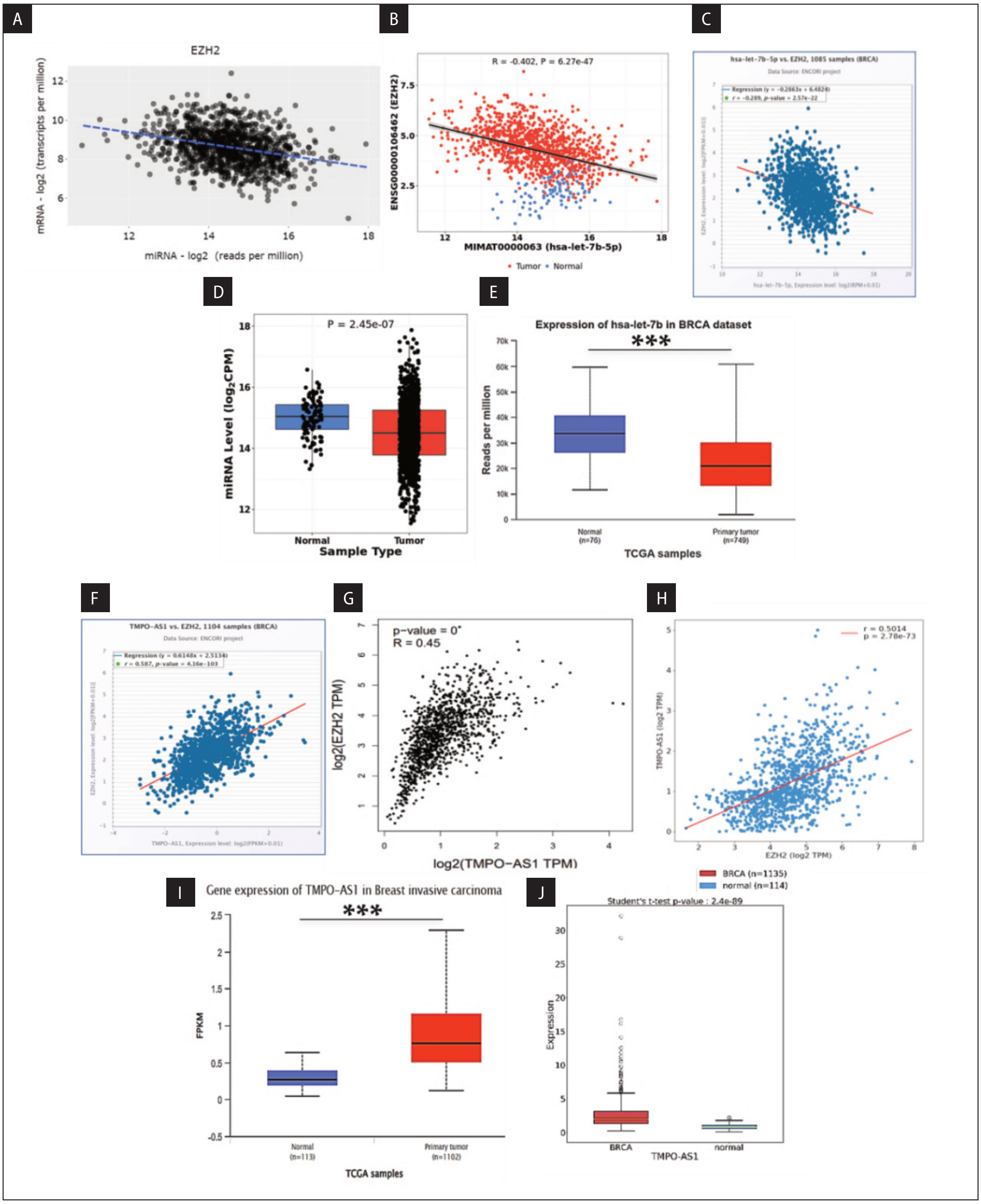

Figure 8. ceRNA network analysis with EZH2 in tumor tissues from breast cancer patients. Correlation analysis between EZH2 and hsa-let-7b-5p using: **A.** TACCO, **B.** CancerMIRNome, and **C.** ENCORI databases. Boxplots showing hsa-let-7b-5p expression in Tumor *vs*. Normal using **D.** CancerMIRNome and **E.** UALCAN in breast cancer (n = 149) *vs* normal (n = 78); **F.–H.** ENCORI , GEPIA and TIMER databases showing correlation between TMPO-AS1 and EZH2; **I.–J.** Boxplots of TMPO-AS1 gene expression in Normal *vs*. Tumor using UALCAN and OncoDB database

Table 2. Correlation of enhancer of zeste homolog 2 (EZH2) *vs.* long non-codin ribonucleic acid (lncRNAs)

Index	Name	Correlation value with EZH2	p-value
1	TMPO-AS1	0.587	4.16e–103
2	PRC1-AS1	0.335	2.19e–30
3	LINC01775	0.350	4.07e–33
4	DEPDC1-AS1	0.480	1.12e–64

because lncRNAs are stable and can regulate miRNA expression. Very interestingly, TMPO-AS1 *vs.* ESR1 (R = –0.167, p = 2.41e–08) and TMPO-AS1 *vs.* PGR (R = –0.232, p = 5.52e–15) also showed a negative correlation, indicating the role of TMPO-AS1 in tumor aggression. The results are shown in Supplementary File — Fig. S5A–D. Also, a network created using the miRNet database between mRNA-miRNA-lncRNA shows direct interaction between EZH2-hsa-let-7b-5p-TMPO-AS1, as shown in Supplementary File — Fig. S5E.

Discussion

EZH2, an essential gene involved in breast cancer development and progression, regulates gene expression and chromatin structure. Understanding EZH2's role could lead to the development of targeted therapies that improve treatment options for patients. Preclinical studies have demonstrated the potential of EZH2 inhibitors for targeting and inhibiting EZH2, a protein frequently overexpressed in breast cancer cells [42, 43]. We have verified the table from the GSCA database and have already explained the relation of positive and negative correlation between drugs and EZH2. The positive correlation between the gene and the drug could lead to drug resistance, whereas the negative correlation could make the drug sensitive. Challenges in targeting EZH2 as a therapy include potential resistance mechanisms, the need to identify biomarkers for patient response, and long-term effects on normal cells and tissues. Small molecular inhibitors or gene knockdowns inhibit EZH2, which reduces cancer cell growth and tumor formation. Recent studies suggest targeting histone methylation as a promising approach for cancer therapy. A bioinformatic study of 2,000 breast cancer patients showed that EZH2 HMT is a major epigenetic driver of TNBC progression. Patients with TNBC had higher levels of EZH2 expression than patients who did not have TNBC [44]. Screening predictive biomarkers, such as EZH2 mutation or overexpression, is important for personalized, precision therapy. Combining EZH2 inhibitors with other treatments is critical, as preclinical studies have shown that EZH2 inhibitors combined with immunotherapy or chemotherapy have a synergistic effect.

The study reveals that EZH2's peak time is in the S phase and is associated with quiescence, invasion, and hypoxia in breast cancer progression. Previous study has shown EZH2 to be a key component of the polycomb repressive complex 2 (PRC2), and is known to methylate histone H3 on lysine 27 (H3K27me3), leading to gene silencing, and this epigenetic modification is crucial for regulating the expression of various cell cycle checkpoints involved in cell cycle progression [45–47]. For example, Tazemetostat is a Food and Drug administration (FDA)-approved drug acting as an EZH2 methyltransferase inhibitor, working on the same mechanism. Targeting EZh2's role in cell proliferation can be approached by disrupting its interaction with key cell cycle regulators. The study found a strong positive association between EZH2 and Cyclin/CDKs, suggesting that EZH2 promotes the G1/S transition, leading to DNA replication, and inhibiting it may cause cell cycle arrest in the G1 phase. Additionally, EZH2 is crucial for the G2/M transition, preventing cancer cells from entering mitosis and reducing tumor cell proliferation. This study also revealed the EZH2 DNA hypomethylation. To our knowledge, this is among the first study on EZH2 DNA hypomethylation in BRCA cases. During the development of cancer, DNA demethylation may involve hemimethylated dyads as bridges, and then the loss of methylation spreads to both strands. The study linked higher levels of EZH2 transcripts to worse RFS, OS, DMFS, and PPS for all breast cancer patients. The study found a strong link between EZH2 and the transcription factor E2F2, which promotes breast cancer by increasing gene expression that promotes cell proliferation and survival while suppressing genes that inhibit these processes. Cancer controls EZH2 gene expression, leading to aggressive tumor growth. Targeting EZH2 in breast cancer treatment holds enormous promise as a potential therapeutic strategy. The study also found that only seven out of 24 miRNAs had a negative correlation with EZH2 gene expression. In BRCA patients, hsa-let-7b-5p had the strongest negative correlation with EZH2. BRCA patients significantly overexpressed TMPO-AS1, which had the strongest positive correlation with EZH2. Further research should explore the potential clinical applications of manipulating EZH2 regulation in cancer treatment. Targeted therapies focusing on the molecular causes of prostate cancer, ovarian cancer, and lung cancer, all linked to abnormal EZH2 expression, could potentially treat these diseases.

Conclusion

The study links the hsa-let-7b-5p-TMPO-AS1-EZH2 Mediated Network, a key epigenetic modulator in TNBC subtypes, to poor overall survival. When analyzed at the molecular level, EZH2 showed that metastasis spreads faster. Understanding subtype-specific regulation is crucial for developing targeted therapies and personalized treatment approaches. It's possible that blocking TMPO-AS1-EZH2 could lead to the development of TMPO-AS1-EZH2 as a targeted treatment for metastases from TNBC.

Conflicts of interest

The authors declare that they have no competing interests.

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None declared.

Author contribution

R.N.: conception, study design, critical reading, and intellectual assessment of the manuscript, study design, and preparation of the manuscript. PV: study design, and preparation of the manuscript. BB: critical reading. SN: critical reading. AS: study design, and preparation of the manuscript.

Data availability

The data for this in-silico study were sourced from publicly accessible databases. The respective links and references for these datasets are provided in the methodology section for transparency and reproducibility.

Ethical clearance

Since this study exclusively utilized publicly available online databases for data extraction and analysis, ethical approval was not necessary for the preparation of this article.

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